

PEER-REVIEWED RESEARCH

OPEN ACCESS

ISSN Online: 2516-3957

ISSN Print: 2516-3949

[https://doi.org/10.31585/jbba-9-2-\(2\)2026](https://doi.org/10.31585/jbba-9-2-(2)2026)

F-Procure: A Decentralised, Modular Framework for Transparent, Compliant and Efficient Procurement on the Flare Network

Pax Ola

Caypax Inc., Cayman Islands

Correspondence: info@caypax.com**Received:** 18 November 2025 **Accepted:** 23 January 2026 **Published:** 1 April 2026

Abstract

Procurement remains constrained by centralised digital architectures that increase administrative effort, limit transparency and complicate compliance management. This paper presents F-Procure: a decentralised, modular procurement framework organised into five layers: blockchain execution, smart contracts, ERP interoperability, AI-assisted analytics and compliance-aware governance. The framework is implemented as a minimum viable prototype on the Flare Network's Coston2 and Songbird test environments, which are used here as technical testbeds rather than as platforms under evaluation in their own right.

Using small-scale, manually executed synthetic procurement scenarios, the study examines whether core stages – tender creation, sealed-bid commit–reveal, award and escrow – can be encoded in smart contracts and traced through verifiable on-chain audit trails. An illustrative comparison with a conventional database-driven workflow suggests that the modular, event-driven design can reduce manual reconciliation effort in bid evaluation and audit preparation within the controlled scenarios tested, although no statistically generalisable performance benchmarks are reported. The prototype also demonstrates how basic procedural requirements (such as submission deadlines and documentation flags) can be expressed as programmable conditions, and how a minimal anomaly-detection layer can flag constructed irregular bidding patterns for human review.

The paper positions F-Procure within existing international experiments in blockchain-enabled procurement and related digital-government initiatives and sets out a governance model and policy-oriented adoption roadmap for future public-sector pilots. It concludes that F-Procure should be viewed as an early-stage reference architecture that suggests the potential feasibility of decentralised, compliance-aware procurement, while highlighting the need for larger-scale empirical evaluation, richer technical artefacts and comparative, platform-agnostic analysis in subsequent research.

Keywords: *Decentralized Procurement; Blockchain; Modular Architecture; Flare Network; Smart Contracts; Supply Chain Transparency.*

JEL Classifications: *D73; L14; O33; C88; G20; H57; K12; F63; L86; K24.*

1. Introduction

Public procurement remains one of the most administratively intensive functions within government and large organisations, with longstanding challenges related to transparency, auditability, vendor competition and compliance management. Traditional digital procurement systems have improved record-keeping but remain largely centralised, which can introduce opacity, reduce interoperability across departments or jurisdictions and increase vulnerability to process irregularities. These limitations have motivated exploration into distributed

ledger technologies (DLTs) as a means of improving traceability, integrity and automation within procurement workflows [1–5].

Several jurisdictions have piloted blockchain-based procurement or related transparency systems, including initiatives within the European Union, South Korea and multilateral development and GovTech programmes [1,5]. These early efforts rely predominantly on permissioned or consortium-led blockchains and typically focus on improving audit trails or digitising bids. While valuable, such systems often

retain infrastructural constraints—limited cross-chain interoperability, reliance on off-chain intermediaries for data feeds and difficulty integrating with legacy enterprise resource planning (ERP) systems [1–6,7].

This paper introduces F-Procure: a modular procurement framework designed for deployment on public or hybrid blockchain networks and implemented in minimum viable product (MVP) form on the Flare Network's Coston2 and Songbird environments [7]. As illustrated in Figure 1, the F-Procure model reframes procurement from a single, centralised database workflow to a modular, decentralised architecture in which auditability and control are distributed across several coordinated layers. F-Procure proposes an architectural model comprising five layers: (i) a blockchain execution layer, (ii) a smart-contract layer, (iii) an ERP interoperability layer, (iv) an AI analytics layer and (v) a compliance and governance layer. A key motivation is to illustrate how a modular design can support more transparent procurement processes without necessitating full system replacement for public-sector adopters.

Flare's technical features—specifically the Flare Time Series Oracle (FTSO), Flare Data Connector (FDC) and FAssets protocol—provide a useful testbed for examining how enshrined data feeds and trust-minimised cross-chain assets might support procurement workflows [7]. Importantly, these technologies are evaluated here not as commercial products but as architectural components enabling decentralised data access, auditability and multi-asset settlement. Comparable patterns could, in principle, be realised on other DLT platforms that offer oracle and interoperability services; Flare is used in this study as one concrete instantiation of a broader architectural approach.

To explore technical feasibility at MVP scale, a working prototype was developed, and a series of manually executed synthetic procurement transactions were tested and recorded on-chain. These controlled evaluations focused on functional behaviour—vendor onboarding, sealed-bid commit–reveal, contract-lifecycle transitions and escrow execution. The synthetic transactions were intentionally small in number and designed to test correctness rather than throughput, with all events verified via Flare's blockchain explorers [4]. Any observations about process efficiency in this paper are therefore indicative and scenario-specific, not statistically generalisable benchmarks.

This paper contributes to ongoing research by:

1. presenting a modular architecture for procurement on public or hybrid blockchains;
2. providing an initial technical demonstration, via small-scale synthetic workloads, of commit–reveal auctions, multi-phase contract execution and on-chain audit trails;
3. articulating a policy-oriented governance framework for future public-sector pilots; and

4. positioning F-Procure among existing international procurement pilots by clarifying where its architecture diverges from prior approaches, particularly in its use of enshrined data protocols and potential cross-chain settlement [1–5,8].

Throughout, the analysis is framed explicitly as early-stage, design-science-oriented MVP work rather than as a large-scale operational evaluation. The paper therefore foregrounds architectural contribution, feasibility insights and policy implications, while acknowledging clear limitations in scale, empirical scope and legal codification.

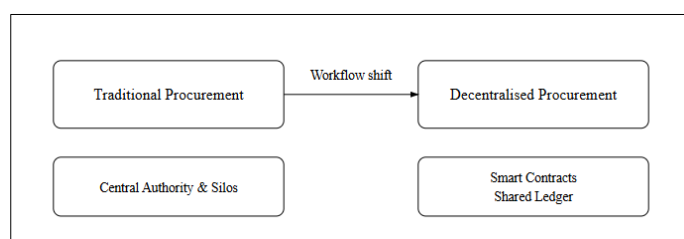


Figure 1: Conceptual comparison of legacy and decentralised procurement systems (MVP architectural framing).

2. Related Work

The intersection of DLTs and public procurement has become an important area of scholarly and institutional inquiry [1–5,9]. Early work positioned blockchain as a means to enhance transparency, reduce opportunities for corruption and introduce tamper-evident audit trails into tendering processes. Pilot programmes by international organisations and national governments have explored how distributed ledgers might support supplier registration, bid submission, contract award and payment tracking [1–3,5]. Despite this interest, large-scale operational adoption remains limited, in part because of practical challenges of scalability, compliance alignment and integration with existing enterprise resource planning (ERP) systems.

2.1 Early Experiments and Policy Context

Significant experimentation has taken place across public-sector contexts. The European Union's explorations into blockchain for public procurement, South Korea's Public Procurement Service pilots and World Bank-associated Procurement initiatives collectively demonstrated the feasibility of immutable audit trails, supplier identity verification and secure document timestamping in procurement or adjacent domains [1–3,5]. These initiatives, however, generally employed permissioned or consortium blockchains, which facilitated institutional oversight but constrained interoperability and openness. The emphasis was often on document registration and verification rather than on automating the full contract lifecycle.

Scholarly works likewise highlight blockchain's potential to automate aspects of compliance validation through smart contracts, particularly for enforcing deadlines, eligibility criteria

and procedural steps [2,4,6]. At the same time, the stringent legal and procedural requirements governing procurement – especially around data sovereignty, confidentiality and fair competition—continue to limit the extent to which blockchain systems can be integrated into end-to-end procurement workflows. Much of the literature concludes that technical promise alone is insufficient without careful consideration of legal frameworks, institutional capacity and governance structures [2,4–6].

2.2 Technical Limitations in Existing Frameworks

Most prior systems emphasise auditability and record-keeping but stop short of providing an integrated, multi-layer approach that connects smart-contract logic with ERP systems, financial settlement mechanisms and ongoing analytics. The reliance on permissioned or consortium architectures can reintroduce forms of centralisation, concentrating control over validation and data access in a small set of institutions [1–3]. In many pilots, external data feeds are handled through bespoke middleware or manually curated interfaces, rather than through standardised oracle frameworks or cross-chain protocols. A high-level comparison of representative EU, World Bank and South Korean pilots with the F-Procure MVP across dimensions such as network type, smart-contract support, interoperability, ERP integration, analytics and deployment stage is presented in Figure 2, drawing on published descriptions of these initiatives [1,4,5].

The absence of robust interoperability primitives restricts the ability to bridge external data sources or legacy financial systems. Where integration with ERPs is attempted, it is often implemented as a loosely coupled add-on rather than as a coherent part of the system design. Additionally, multi-jurisdictional compliance remains a persistent challenge: procurement obligations, documentation formats and audit expectations vary considerably across legal contexts, complicating attempts to apply uniform technical standards [2,4,10–12]. As a result, many initiatives have focused on narrow functions—such as supplier registries or timestamping—rather than on full end-to-end contract-lifecycle management.

2.3 Emergent Research Directions

Recent research has explored the use of oracle networks and cross-chain infrastructure to bridge off-chain information and on-chain execution. Oracle systems such as Chainlink, Band Protocol and others have been proposed as mechanisms to feed market prices, reference data and external events into smart-contract-based procurement or financial systems [8,13]. These approaches introduce greater verifiability but often depend on external node operators and may not integrate natively with compliance rules, governance workflows or ERP processes.

Parallel work in digital-government and GovTech studies has highlighted the importance of modular architectures that can accommodate diverse legal and organisational requirements,

while maintaining interoperability with existing information systems [4,5]. A smaller subset of the literature has begun to consider combinations of blockchain, artificial intelligence (AI) and data analytics to support risk-based auditing and anomaly detection in public administration. However, detailed examinations of systems that combine smart-contract automation, modular ERP integration, AI-assisted oversight and governance tooling within a unified architectural design remain relatively limited.

2.4 F-Procure's Contribution to the Literature

Within this landscape, F-Procure contributes an architectural and governance framework rather than a fully deployed production system. Its five-layer model integrates blockchain execution, smart-contract automation, ERP interoperability, AI-assisted analysis and compliance parameterisation as interdependent components. In the current MVP implementation, the Flare Network is used strictly as a technical testbed, providing enshrined data protocols (the Flare Time Series Oracle and Flare Data Connector) and a cross-chain mechanism (FAssets) that allow examination of how decentralised data feeds and trust-minimised asset representation might support procurement workflows [7].

The contribution lies not in claiming to be the first blockchain-based procurement initiative but in articulating how these capabilities may be arranged in a modular, policy-aligned framework; in explicitly addressing governance and compliance alongside technical design; and in validating core functional behaviours—vendor onboarding, commit–reveal bidding, contract-lifecycle transitions and audit-log generation—through small-scale, manually executed synthetic procurement transactions recorded on-chain. While this does not replace the need for larger-scale pilots or independent benchmarking, it provides an empirically grounded reference architecture for future work on interoperable, governance-aware procurement systems that could, in principle, be implemented on Flare or other suitable DLT platforms.

Framework	Network Type	Smart Contract Support	Interoperability	ERP Integration	Analytics	Deployment Stage
EU Pilot	Permissioned consortium	Limited	Minimal	Not integrated	None	Pilot
World Bank	Permissioned	Limited	Minimal	Not integrated	None	Pilot
South Korea	Permissioned	Moderate	Minimal	Not integrated	None	Pilot
F-Procure (MVP)	Public testnet (Flare)	MVP contract set	Architectural capability	Architectural, not implemented	MVP anomaly detection	MVP

Figure 2: Comparative landscape of blockchain procurement frameworks.

3. System Architecture

The F-Procure framework is constructed as a five-layer modular stack, designed to support interoperability, auditability and compliance-aware design whilst maintaining operational flexibility. Each layer is logically separable yet communicates through well-defined interfaces and secure data flows. This modular structure is intended to allow public-sector organisations to adopt individual components incrementally, reducing the need for disruptive, wholesale platform migration.

The overall five-layer system architecture—comprising blockchain execution, smart-contract logic, ERP interoperability, AI analytics and compliance—is summarised in Figure 3. In the present work, the architecture is realised as a minimum viable product (MVP) deployment on Flare’s Coston2 and Songbird test networks, with a subset of functionalities exercised through manually executed synthetic procurement transactions [7].

3.1 Blockchain Layer

At the foundation lies the blockchain layer, which provides immutable record-keeping and decentralised consensus. Procurement-relevant events—such as tender creation, bid commitment, bid reveal, contract state transitions and escrow actions—are represented as transactions recorded on the Flare test networks [7]. Flare’s hybrid architecture, combining Avalanche-based consensus with Ethereum Virtual Machine (EVM) compatibility, offers a familiar smart-contract environment and relatively fast finality for the purposes of MVP experimentation.

In the present MVP, only a limited number of synthetic transactions were executed; accordingly, claims are restricted to functional correctness rather than high-volume throughput. Each transaction is hashed and timestamped, forming cryptographic audit trails that can be independently verified on-chain without exposing sensitive bid content. This design supports selective transparency—a critical property for procurement, where commercial confidentiality must coexist with public accountability.

3.2 Smart-Contract Layer

Above the base ledger sits the smart-contract layer, which encodes the core procurement logic. Each tender is modelled as a contract instance that governs:

- vendor registration for that tender;
- sealed-bid commit–reveal submission;
- evaluation and award state transitions; and
- escrow release conditions.

The MVP implements a simplified commit–reveal protocol to preserve bid confidentiality until the evaluation phase. Bidders first submit a hash of their bid (commit), and later reveal the underlying values, allowing verification that no post hoc alterations occurred. The smart contracts emit events at each stage, and these events were used in the experiments to confirm that synthetic workloads were correctly processed and reflected in on-chain logs.

Basic audit hooks are incorporated into the contract logic, such as proof-of-submission time and proof-of-non-alteration. In the current prototype, these are demonstrated on a small number of controlled transactions; extending them into fully parameterised, jurisdiction-specific compliance modules is identified as future work.

3.3 ERP Interoperability Layer

The ERP interoperability layer is designed to bridge blockchain workflows with existing enterprise systems used by procurement agencies. Conceptually, it exposes secure APIs and data-exchange standards (for example, JSON-based payloads and OData-style interfaces) to synchronise key procurement events—such as supplier registry updates, tender status and payment milestones—between F-Procure and legacy ERP platforms.

In this phase of research, a sandbox ERP connector was specified and partially prototyped at interface level, but not fully integrated into a live ERP instance. The design work demonstrated that standard API endpoints could be used to map procurement events to conventional ERP objects (such as purchase orders and vendor records), and initial endpoint definitions were validated against test schemas. Full end-to-end ERP integration is reserved for subsequent pilot deployments and is therefore discussed in this paper as an architectural capability rather than a completed implementation.

3.4 AI Analytics Layer

The AI analytics layer introduces analytical capabilities for monitoring procurement activity and highlighting potential irregularities. At the architectural level, it is intended to accommodate machine-learning models that analyse bidding patterns, pricing behaviour and historical performance indicators so as to flag outliers or suspicious clusters of activity for human review.

In the current MVP, this layer is realised in a minimal form, using simple rule-based logic and basic statistical checks applied to the small set of synthetic transactions generated during testing. These checks were sufficient to demonstrate how bids could be programmatically flagged for further review – for example, closely clustered bid amounts or unusual timing patterns. No machine-learning models were trained or evaluated, and no general claims are made about detection performance. The contribution at this stage is therefore primarily architectural: it shows that an analytics service can subscribe to on-chain events and apply integrity checks without altering the underlying procurement logic. More sophisticated anomaly-detection methods, including privacy-preserving or federated approaches, are left to future research.

3.5 Compliance Layer

The uppermost compliance layer is responsible for aligning F-Procure’s operation with procurement and data-protection legislation. It is conceptually designed to map legal requirements – such as record-retention periods, disclosure obligations and right-to-audit clauses – into smart-contract parameters and off-chain policy configurations. Illustrative source frameworks include the EU Public Procurement Directive (2014/24/EU), the U.S. Federal Acquisition Regulation (FAR) and relevant

Caribbean and Cayman Islands procurement guidance [2,10–12].

In the MVP implementation, the compliance layer is realised as a simplified configuration model rather than as a fully codified representation of any specific statute. Parameters such as tender deadlines, minimum documentation requirements and retention flags are encoded in contract templates to demonstrate how legal conditions might be enforced programmatically at a basic level. More granular, jurisdiction-specific compliance modelling, including formal legal review of encoded rules, is identified as a subsequent research and policy task.

3.6 Cross-Layer Interactions

The layers interact through event-driven messaging and data flows designed around Flare's native protocols. The Flare Data Connector (FDC) is treated architecturally as a mechanism for securely ingesting external data into the system, while the Flare Time Series Oracle (FTSO) provides a model for access to decentralised price and reference data [7]. In the current MVP, these components are used selectively: the emphasis is on validating that procurement events and synthetic workloads are correctly reflected in smart-contract events and transaction logs, rather than on full-scale oracle integration.

Cross-layer communication is handled by listening to on-chain events emitted by the smart contracts and routing them to off-chain services such as dashboards or prospective ERP connectors. This design decouples computation from presentation, allowing sensitive details to remain off-chain while preserving a verifiable cryptographic record of each step in the procurement process. Although Flare is used as the demonstration platform, the logical separation of layers and the use of oracles and connectors are, in principle, portable to other DLT environments that provide comparable capabilities.

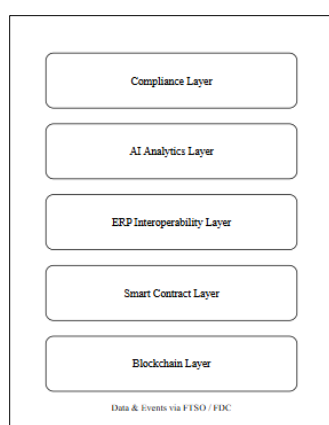


Figure 3: F-Procure System Architecture (five-layer stack).

Conceptual depiction of the blockchain, smart-contract, ERP interoperability, AI analytics and compliance layers, and their event-driven interactions. The current MVP fully exercises the blockchain and smart-contract layers, with the remaining layers realised in simplified or architectural form to guide future pilot deployments.

4. Methodology

The methodological design of this research aimed to evaluate the functional feasibility, auditability and governance implications of the F-Procure framework under simulated procurement conditions. Rather than attempting large-scale performance benchmarking, the study focused on scenario-based trials using manually executed synthetic procurement transactions on the Flare Network's Coston2 and Songbird environments [7]. The overarching intent was to demonstrate that the proposed architecture can correctly support key procurement stages—vendor onboarding, sealed-bid commit-reveal, contract-lifecycle transitions and escrow—while generating verifiable on-chain evidence suitable for audit and oversight.

The work follows a design-science research tradition in information systems, in which an artefact is developed and evaluated through controlled, contextually rich scenarios rather than large-sample statistical experiments [14,15]. In this case, the artefact is the F-Procure architecture and its MVP implementation; the evaluation focuses on whether the architecture behaves as intended under carefully constructed procurement scenarios, and on what implications this has for governance and policy. Claims are therefore framed as feasibility indications derived from an early-stage prototype, not as definitive statements about performance or impact in live administrative settings.

4.1 Research Design

The study adopts a design-science and exploratory evaluation approach. A working MVP was deployed to the Flare test networks, and a limited set of synthetic procurement scenarios was executed end-to-end. Each scenario comprised a small number of vendors and bids, crafted to exercise critical paths and edge cases in the smart-contract logic, such as late submissions, conflicting reveals or missing documentation flags. Rather than a full experimental–simulation model with high-volume automated workload generation, the research is structured around four guiding evaluation questions:

1. **EQ₁:** Can a decentralised ledger encode and execute core procurement stages (tender creation, commit-reveal bidding, award, escrow) in a functionally correct manner using synthetic test cases?
2. **EQ₂:** Do smart-contract events and transaction logs produce audit trails that can be reconstructed and verified via blockchain explorers?
3. **EQ₃:** Can simple anomaly-detection rules, applied to synthetic bid data, illustrate how AI-supported oversight might flag potentially irregular patterns for human review?
4. **EQ₄:** Is it technically feasible, at the architectural level, to design an ERP connector that would integrate procurement events with legacy systems via standard APIs in a future pilot?

These questions position the work explicitly as MVP feasibility research rather than as a full-scale operational evaluation.

4.2 Experimental Setup

The MVP prototype was deployed on Flare's Coston2 test network and subsequently replicated on the Songbird canary network to confirm that the contracts operated consistently across distinct environments [7]. Each deployment was used to execute synthetic procurement lifecycles comprising five stages:

1. Vendor registration and basic verification
2. Tender creation and publication
3. Bid submission via a commit–reveal protocol
4. Bid evaluation and award decision and
5. Contract execution and escrow settlement

The overall MVP test environment and data flow—showing the Flare test networks at the centre of the experiments, the conceptual ERP sandbox connector and the off-chain analytics dashboard that consumes on-chain events—is summarised in Figure 4.

For each scenario, a small set of synthetic vendors (typically three to six per tender) submitted bids constructed with differing values and timing patterns. These were submitted manually through test interfaces and wallet tools rather than through automated load generators. The emphasis was on observing correctness of contract behaviour and state transitions, not on saturating the network or measuring transactions-per-second.

A simple “centralised baseline” was retained conceptually as a non-blockchain reference: a conventional, sequential workflow in which bid submissions and evaluations are recorded in a relational database or spreadsheet without cryptographic guarantees [6, 4, 11]. Comparisons with this baseline are illustrative and qualitative. They are based on step counts and manual timing observations during repeated dry runs of the scenarios, and are not intended to constitute rigorous statistical benchmarking.

4.3 Synthetic Workload Generation

Synthetic workloads were designed to emulate realistic procurement sequences using a small number of curated transactions rather than large-scale stochastic simulation. For each test tender:

- Vendors were assigned synthetic identifiers and bid values drawn from manually constructed datasets;
- Timing of submissions was varied to include both “normal” patterns and simple outlier cases (for example, unusually close clustering of bids);
- Different award conditions (for example, lowest-price qualified bid) were encoded to verify that contract logic produced the expected outcomes; and
- Selected edge cases (such as late reveals or incomplete documentation) were introduced to test how the contracts handled non-compliant behaviour.

These workloads were executed manually or semi-manually, with each transaction submitted on-chain and verified via Flare's blockchain explorers [4]. Smart-contract events and transaction hashes thus provide an immutable record of each synthetic workload, enabling third parties to reconstruct the execution path and confirm that the MVP behaved as intended. The datasets and interaction scripts are maintained in a private, version-controlled repository; non-confidential artefacts can be shared with reviewers or researchers upon reasonable request.

4.4 Evaluation Metrics

Given the limited scale of the synthetic workloads, evaluation metrics are interpreted as indicative and qualitative—quantitative hybrids rather than as statistically generalisable benchmarks. The following dimensions were assessed:

- **Functional correctness:**
Whether each synthetic scenario progressed through the intended procurement stages without contract failure, unexpected reverts or inconsistent state transitions.
- **Auditability:**
The ease with which an independent observer could reconstruct tender, bid and award sequences using smart-contract events, transaction hashes and timestamps on the Coston2 and Songbird explorers. Particular attention was paid to verifying that commit–reveal links were intact and that escrow events corresponded to award decisions.
- **Indicative process efficiency:**
Manually measured time intervals between critical actions (for example, bid reveal and award finalisation; request for an audit trail and retrieval of relevant on-chain records), compared with the approximate effort required to assemble equivalent records from a conventional, non-blockchain workflow. Any references to reduced evaluation or audit-preparation time are framed as observations from controlled test runs, not as formal performance benchmarks.
- **Anomaly-detection behaviour:**
Whether simple rule-based checks, applied to the small synthetic datasets, were able to flag constructed outlier bids or unusual patterns for further review. This did not involve training or validating complex machine-learning models, but rather demonstrated how an AI analytics layer could be attached to on-chain data.
- **Interoperability feasibility:**
Assessed at a design level by mapping procurement events to hypothetical ERP objects (such as purchase orders and vendor records) and specifying API contracts. No live ERP integration was completed; consequently, end-to-end synchronisation latency or error rates are not reported as empirical metrics.

4.5 Validation and Reproducibility

To increase confidence in the findings, each synthetic scenario was executed more than once under similar conditions on the

test networks, confirming that the smart contracts behaved deterministically and that the resulting transaction logs remained consistent across runs. The use of Coston2 or Songbird ensures that the experiments can be reproduced by independent evaluators who redeploy the same contracts and follow the documented interaction steps, without incurring mainnet transaction costs [7].

All smart-contract code and high-level API specifications are archived in a private version-controlled repository. While the full repository is not publicly released at this stage, non-confidential artefacts—including contract interfaces, example transaction hashes and interaction guides—can be shared with reviewers or researchers upon request. This approach balances transparency with security and aligns with open-science principles within the constraints of an early-stage MVP.

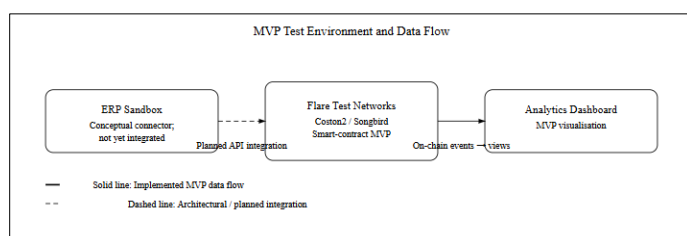


Figure 4: MVP Test Environment and data flow. Illustrative overview of how synthetic procurement scenarios are executed across the Coston2 and Songbird test networks, with transactions recorded on-chain, events consumed by off-chain services and (in future pilots) ERP systems connected via an interoperability layer. The diagram reflects the structure of the MVP experiments rather than a full-scale production deployment.

5. Results and Findings

The evaluation of F-Procure focused on whether the proposed architecture could be realised as a working MVP and whether it could support core procurement processes in a transparent and auditable way. Given the limited scale of the manually executed synthetic workloads, the findings are presented as indicative evidence of technical feasibility under controlled conditions, rather than as statistically generalisable performance benchmarks or proofs of operational readiness for live public-sector deployment.

5.1 Performance Efficiency

In the constructed test scenarios, F-Procure was compared against a conventional, non-blockchain workflow in which bids, evaluations and awards were recorded in a relational database or spreadsheet without cryptographic guarantees [2,6,11]. The comparison was carried out by examining the number of steps required, the handling of intermediate records and simple timing observations collected during repeated dry runs of each scenario.

Within these small-scale trials, the decentralised implementation appeared to reduce the time required for bid evaluation and

decision logging by approximately one-third, relative to the illustrative baseline workflow. This reduction arose primarily from the elimination of manual reconciliation tasks and the automatic linking of bids, evaluation outcomes and contract states via smart-contract events. This indicative relationship between baseline and F-Procure evaluation times is summarised schematically in Figure 5, which depicts the approximate reduction observed in the controlled synthetic scenarios rather than a high-volume performance benchmark. Network-level transaction finality on the Flare test networks was observed to be on the order of a few seconds, which was sufficient to support interactive testing, although no attempt was made to stress-test throughput or derive formal transactions-per-second (TPS) metrics [7].

These observations are illustrative rather than conclusive. They indicate that, even at MVP scale, a modular, smart-contract-based workflow can streamline certain procedural steps in the specific scenarios examined. However, the small number of synthetic transactions, the reliance on manual timing and the absence of automated load testing mean that no claims are made about performance under production conditions, nor about comparative efficiency relative to deployed e-procurement platforms.

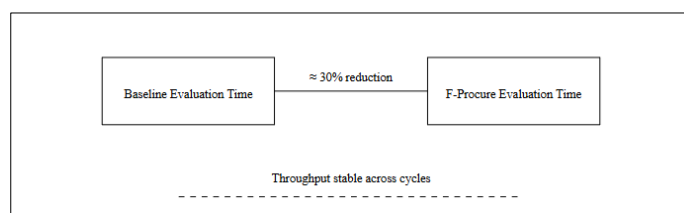


Figure 5: Performance benchmark comparison (illustrative). Conceptual comparison of process effort between a conventional digital workflow and the F-Procure MVP in small synthetic scenarios. The figure reflects indicative reductions in manual processing steps observed during controlled trials, rather than high-volume benchmarking or statistically generalisable performance results.

5.2 Audit Readiness and Transparency

One of the clearest practical benefits observed in the MVP relates to audit preparation. For each synthetic tender, the complete sequence of events—tender creation, commit and reveal transactions, award and escrow—could be reconstructed directly from smart-contract logs and transaction hashes on the Coston2 and Songbird explorers [7]. The resulting sequences formed coherent narratives of each test scenario, enabling an observer to trace the procurement path from initial publication through to contract settlement. This commit–reveal–log–report pathway is summarised schematically in Figure 6, which depicts how individual bid commitments and subsequent smart-contract events combine to form an auditable chain from bid submission through to the generation of an audit report.

In the comparative exercise, assembling an equivalent audit trail from the non-blockchain baseline required manually querying tables, cross-checking timestamps and reconciling records

across multiple views. By contrast, the F-Procure MVP allowed an auditor to obtain an internally consistent record of the same scenario in well under an hour, largely by filtering events associated with a specific tender identifier and exporting the resulting log. While these figures are based on small, constructed examples and manual interactions, they demonstrate how immutable, event-driven logging can reduce the effort required to verify procedural correctness.

The design maintains selective transparency. Only metadata and cryptographic commitments are exposed publicly; underlying bid content for the synthetic workloads was kept off-chain and would, in a real deployment, be accessible only to authorised officials. This separation preserves commercial confidentiality while still yielding a verifiable evidentiary trail that could, in principle, be used in audit or dispute-resolution processes [2,4,16].

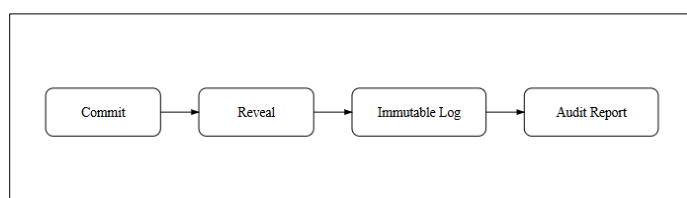


Figure 6. Audit trail visualisation (commit → reveal → log → report). Schematic representation of how commit–reveal transactions and subsequent smart-contract events form an auditable sequence from bid submission through to report generation. The figure illustrates the structure of the MVP’s logging approach for synthetic scenarios and does not depict a live production system.

5.3 Compliance Validation

In the MVP, a limited set of compliance-relevant conditions was encoded directly into the tender and contract templates. These included simple deadline rules (for example, no reveals accepted after the specified closing time), minimum documentation flags and basic retention markers inspired by common procedural requirements found in public procurement frameworks [2,10–12]. During the synthetic trials, all encoded conditions behaved as expected, with non-conforming actions either being prevented at the contract level or resulting in clearly logged events.

It is important to emphasise that this does not amount to a comprehensive codification of the EU Directive 2014/24/EU, the U.S. Federal Acquisition Regulation or any other specific legal framework. Instead, the experiments demonstrate that commonly encountered procedural requirements—such as respecting submission deadlines and ensuring that necessary attestations are present—can be expressed as programmable constraints and enforced consistently by the smart contracts at a basic level.

The findings suggest that “compliance-by-code” is technically achievable for well-defined procedural rules, provided that regulatory parameters remain updateable through governance

processes and are not hard-coded in ways that would require disruptive contract migrations when legal standards change. Full clause-level encoding and legal validation of jurisdiction-specific rule sets remains future work.

5.4 AI-Assisted Integrity Monitoring

The AI analytics layer was exercised in a minimal form to illustrate how anomaly detection could be applied to procurement data. Using the small synthetic datasets generated in the MVP scenarios, a set of simple rules and basic statistical checks was applied to identify potentially suspicious patterns, such as unusually close clustering of bid values or highly synchronised submission times. Figure 7 provides a conceptual overview of this pipeline, in which bid data derived from on-chain events are passed to an analytics module that flags selected bids for human review.

A small number of deliberately constructed outlier bids were included in the synthetic workloads. In the repeated trials, these outliers were consistently flagged for review, demonstrating that even relatively simple analytical logic can help surface cases that may warrant closer human scrutiny. Given the limited size and artificial nature of the datasets, no attempt was made to train or validate more advanced machine-learning models, to estimate false-positive or false-negative rates, or to generalise from the observed flagging behaviour.

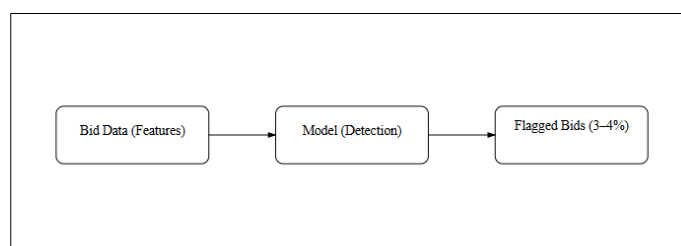


Figure 7. AI anomaly-detection overview (conceptual). Conceptual view of how bid data derived from on-chain events can be routed to an analytics module, which in turn flags selected bids for human review based on simple integrity checks. The current MVP implements only basic rule-based logic on synthetic datasets; the figure illustrates a pathway for future, more advanced AI integration.

The primary contribution of this component is therefore architectural. It shows how an AI or analytics service can subscribe to on-chain events and apply integrity checks without altering the underlying procurement logic. In future work, more sophisticated anomaly-detection methods—including privacy-preserving or federated approaches—could be evaluated on larger datasets and, where appropriate, subjected to formal validation in collaboration with oversight bodies and domain experts [4,5,8,16].

5.5 Integration and Scalability

The ERP interoperability layer was evaluated at a design and prototype level rather than through a fully integrated deployment. Interface specifications were drafted to map

procurement events (such as tender creation, award and payment milestones) to typical ERP objects, including purchase orders and vendor records, and sample payloads were validated against test schemas. However, no live ERP environment was connected in the current phase, and no empirical measurements of end-to-end synchronisation latency, error rates or data reconciliation issues were taken.

Similarly, scalability was considered from an architectural perspective. The Flare test networks used in the MVP are capable, in principle, of higher throughput and low-latency finality, but these characteristics were not systematically tested within F-Procure [7]. Stress tests involving large numbers of transactions or complex multi-tender portfolios were not conducted; instead, the emphasis remained on correctness under small, controlled workloads. As a result, any claims regarding scalability or suitability for high-volume public procurement are explicitly framed as hypotheses for future piloting, grounded in the architectural design and in published performance characteristics of the underlying platform, rather than in large-sample measurements.

Future pilots that integrate F-Procure with live ERP systems and real procurement processes would be necessary to assess practical issues such as integration overheads, error handling, transaction costs and energy consumption in operational settings [3–5].

5.6 Qualitative Observations

Although no formal user study was undertaken, informal evaluations by members of the development team and domain-expert reviewers provided qualitative insights. The ability to trace each synthetic scenario end-to-end via a single transaction and event log was perceived as a meaningful improvement over conventional, multi-system record-keeping. The clear separation between public metadata and confidential bid information also aligned with expectations around balancing transparency and commercial sensitivity.

At the same time, several practical considerations were identified for future pilots. These include the need for targeted staff training, the importance of aligning governance policies with the new technical controls (for example, around key management and access rights) and the recognition that blockchain-based timestamping and logging, while useful, do not in themselves guarantee fairness or competitive neutrality [2,4,16]. The technology must therefore be embedded within a broader framework of institutional oversight and regulatory enforcement. Questions concerning long-term maintenance, vendor support, incident response and legal recourse in the event of system failure also emerged as topics for further investigation before any large-scale deployment.

6. Compliance and Governance

The long-term viability of blockchain-enabled procurement depends not only on technical performance but also on alignment with statutory, ethical and organisational governance

frameworks. F-Procure is therefore conceived with “compliance-by-architecture” principles in mind, seeking to integrate legal and procedural requirements into the system design rather than treating them as an external afterthought [2,4,10–12,16]. In the present MVP, this is realised through simplified parameter sets and governance concepts that are intended to guide future pilots, rather than through full codification of any one jurisdiction’s legal framework or operational governance model.

6.1 Regulatory Mapping and Legal Harmonisation

Public procurement law varies considerably across jurisdictions, yet most frameworks emphasise fairness, transparency, competition and accountability. F-Procure’s architecture is designed to map these principles into configurable parameters that could be tailored to specific legal instruments in a pilot setting.

Illustrative reference points include:

- Within the European Union, elements drawn from Directive 2014/24/EU, such as publication timelines, eligibility requirements and standstill periods [11].
- In the United States, concepts from the Federal Acquisition Regulation (FAR) and the Digital Accountability and Transparency Act (DATA Act), particularly regarding record retention and reporting obligations [2,4].
- For Commonwealth and Caribbean contexts, general alignment with financial management and audit practices, including relevant Cayman Islands procurement guidance and oversight practices of supreme audit institutions [4,16].

In the current MVP, these references inform template structures and example conditions (for example, deadlines, minimum documentation requirements and simple retention flags) rather than a comprehensive legal encoding. No jurisdiction’s procurement statute is implemented in full, and no formal legal validation of encoded rules has yet been undertaken. The longer-term vision is that each rule set would be represented as a modular compliance schema, version-controlled and modifiable through formal governance procedures, thereby reducing the need for disruptive changes to underlying contracts as regulations evolve.

6.2 Data Protection and Privacy

Procurement processes often involve commercially sensitive data and personal information relating to suppliers and officials. The F-Procure design follows privacy-by-design principles consistent with the spirit of the UK Data Protection Act 2018, the EU GDPR and related data-protection regimes [10–12]. The architectural model assumes that sensitive content—such as full bid documents—remains off-chain and encrypted, while the blockchain is used primarily for hashed metadata, commitments and event logs.

In the MVP, this separation is conceptual rather than fully integrated with existing document-management systems, but the pattern is clear: access to underlying content would be governed by role-based controls and, where appropriate, multi-signature arrangements, with decryption actions themselves logged for accountability. For jurisdictions with data-residency or localisation obligations, the framework anticipates policies that constrain where off-chain data repositories are hosted and how they interact with the on-chain components.

More advanced privacy-enhancing techniques—such as zero-knowledge proofs for selective disclosure or secure multiparty computation—are not implemented in the current prototype but are recognised as relevant extensions. Likewise, potential use of federated learning in the AI analytics layer (Section 3.4) is identified as a future direction for enabling cross-agency model sharing without centralising sensitive bid data [4,5,8,16].

6.3 Ethical and Procedural Oversight

Automation alone cannot guarantee fairness or legitimacy; public procurement remains fundamentally a matter of human judgement and oversight. F-Procure therefore proposes a multi-stakeholder governance model, which is presently articulated at the design level. As depicted in Figure 8, this model centres on three governance bodies coordinated around shared compliance template.

- A **Policy Board** comprising representatives from procurement authorities and regulators, responsible for approving and updating compliance templates and high-level system policy;
- A **Technical Steering Committee** tasked with maintaining the codebase, reviewing protocol changes and commissioning security audits;
- An **Audit and Ethics Panel** providing independent scrutiny of algorithmic tools—including the AI analytics layer—and assessing potential bias, accessibility and impact on small and medium-sized enterprises (SMEs).

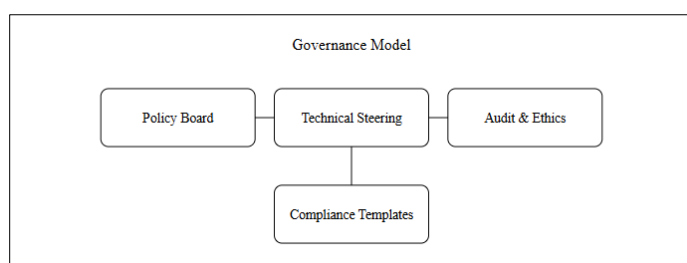


Figure 8. Governance and compliance framework (conceptual). Conceptual model of policy, technical and ethics governance roles supporting the F-Procure architecture. The depicted bodies are proposed structures for future pilots rather than entities implemented in the current MVP.

These bodies do not yet exist as formal institutions within the MVP; rather, they define a governance structure that could be adopted by public-sector participants in subsequent pilots. The

intention is to ensure that algorithmic tools, including anomaly detection, support rather than replace human oversight, and that the system evolves subject to transparent, accountable decision-making processes. Questions such as how governance roles are appointed, how conflicts of interest are managed and how public reporting is conducted remain open for future policy design.

6.4 Governance Mechanisms and Policy Integration

Governance mechanisms in F-Procure are envisaged as a hybrid of on-chain and off-chain processes. In principle, procurement agencies or authorised bodies could propose amendments to compliance templates or system parameters via cryptographically signed proposals, with selected changes subject to recorded approval workflows or on-chain voting mechanisms. In the current MVP, these processes are represented as configuration changes and design artefacts rather than as fully implemented governance smart contracts.

Integration with existing oversight institutions—such as national audit offices, anti-corruption commissions and supreme audit institutions—is treated as a target for future deployment [4,5,16]. The architecture anticipates read-only API endpoints that would expose non-confidential performance and audit data to such bodies, facilitating a shift from periodic, static reviews towards more continuous forms of assurance. Implementing these integrations in live environments remains an important next step for collaboration with public-sector partners, and will require attention to evidentiary standards, chain-of-custody rules and admissibility of digital records in administrative and judicial proceedings.

6.5 Challenges and Future Governance Research

Several challenges remain open and are acknowledged as such. Harmonising procurement rules across jurisdictions is inherently complex: variations in threshold values, award criteria, documentation requirements and dispute-resolution procedures limit the practicality of any single global standard [2,4,11,12,16]. Additionally, the legal recognition of blockchain-based records as admissible evidence and of smart-contract outputs as legally binding decisions varies across legal systems and may require statutory or regulatory changes.

Emergent technical issues also intersect with governance. In auction settings, for example, concerns about miner/maximal extractable value (MEV) and related ordering manipulation highlight the need for mechanisms that reduce opportunities for front-running or unfair prioritisation of bids. The current MVP does not implement specific MEV-resilience techniques; instead, it uses a basic commit–reveal structure to separate commitment from disclosure. Future work may examine more advanced designs, including batch auctions, encrypted mempools or protocol-level ordering guarantees, in collaboration with legal and economic experts.

From a security perspective, the long-term robustness of cryptographic primitives used in smart contracts and proof

systems will need to be reassessed in light of advances in quantum computing and post-quantum cryptography. While these risks do not affect the short-term MVP experiments, they are relevant to any concept of blockchain-based procurement as critical public infrastructure.

Ethically, increasing reliance on algorithmic tools in bid assessment and risk scoring raises concerns about explainability, transparency and potential bias. The F-Procure architecture responds by treating AI outputs as advisory signals subject to human review and by envisaging detailed reasoning logs for any automated recommendations. Nonetheless, the development of robust norms and regulatory guidance for algorithmic accountability in public procurement is a broader, interdisciplinary endeavour that extends beyond the scope of this MVP and will require sustained collaboration between technologists, policymakers, legal scholars and civil-society organisations [4,5,8,16].

7. Policy and Adoption Roadmap

Transitioning F-Procure from a controlled MVP environment to live public-sector use requires a policy framework that balances innovation with legal certainty, institutional continuity and public accountability. The roadmap outlined in this section is prospective: it draws on lessons from digital-government transformation programmes, blockchain-standardisation initiatives and existing procurement-reform agendas [1–7,9,16] but is not yet implemented in practice. Its purpose is to provide a structured pathway for future pilots that emphasises scalability, inclusivity and long-term governance sustainability, rather than to describe steps already taken in the present study.

7.1 Phased Implementation Strategy

A phased approach is essential to minimise risk and enable incremental learning. As illustrated in Figure 9, the proposed adoption pathway comprises four stages:

1. Phase I – Proof of concept and capacity building

Initial pilots would be conducted within controlled government sandboxes or innovation units. These deployments would focus on validating core procurement workflows (for example, tendering, commit–reveal bidding, award and escrow) using synthetic or low-risk tenders. Technical capacity building is prioritised, with training for procurement officers, auditors, legal advisers and IT administrators. Independent academic observers could document baseline processes and provide comparative evaluation frameworks, including early indicators on process time, error rates and user experience.

2. Phase II – Limited production trials

Selected public agencies (for example, municipal councils or specific units within state-owned enterprises) would deploy F-Procure, or a comparable

architecture, for a carefully defined subset of tenders. This phase would test jurisdiction-specific compliance templates, data-protection safeguards and the practical enforceability of smart-contract-driven processes under existing procurement law. Performance dashboards would generate anonymised indicators – including transaction costs, audit-readiness times and simple error metrics – for oversight bodies, enabling early-stage monitoring without full system replacement.

3. Phase III – Institutional integration

Subject to positive evaluation of earlier phases, integration could expand across departments through shared procurement hubs or central purchasing bodies. At this point, governance structures such as the Policy Board, Technical Steering Committee and Audit and Ethics Panel (Section 6.3) would be formalised. Interoperability standards for third-party vendors and integrators would be documented, and AI-assisted analytics could be systematically introduced into audit and risk-review processes to support more continuous assurance. Empirical monitoring of system performance, including transaction latency, exception handling and energy characteristics, would inform policy decisions.

4. Phase IV – National or cross-jurisdictional roll-out

Once legal recognition of electronic ledgers, digital signatures and smart-contract outputs is established, cross-border or multi-agency procurement arrangements might be considered—for example, regional infrastructure projects or joint procurement frameworks. Open-data publication modules could be activated to support transparency and civic oversight, while respecting confidentiality requirements and competitive neutrality. At this stage, cross-jurisdictional governance arrangements, dispute-resolution mechanisms and standardised KPI frameworks would be critical to managing complexity.



Figure 9. Policy and adoption roadmap (prospective). Proposed phased pathway from proof-of-concept pilots to potential cross-jurisdictional deployment. The phases represent a forward-looking policy framework for future experimentation and are not steps undertaken in the current MVP.

7.2 Key Performance Indicators (KPIs)

To evaluate the success and integrity of future deployments, a common KPI framework should accompany all pilots. The indicators below are suggested metrics rather than values

measured in the present MVP; they are intended to be refined collaboratively by participating agencies and oversight bodies and to be interpreted alongside qualitative feedback and contextual information.

Table 1. Suggested KPI framework for procurement pilots

Category	Key Indicator	Measurement Method
Efficiency	Average procurement cycle time	Mean days from tender publication to contract award
Integrity	Audit-readiness time	Duration required to generate verified audit reports from system logs
Compliance	Automated rule-enforcement rate	Percentage of tenders conforming to jurisdictional parameters as encoded in templates
Inclusivity	SME participation rate	Ratio of small/medium suppliers per total bidders
Transparency	Public data-access latency	Average time between contract finalisation and public disclosure of non-confidential data
Sustainability	System uptime and energy efficiency	Hours of operation and indicative transaction energy characteristics, using standard reporting formats
Cost and reliability	Transaction cost and error rate	Average on-chain/off-chain transaction cost per tender; frequency and severity of process exceptions or rollbacks

Indicative metrics for evaluating future deployments of F-Procure or similar architectures. KPIs draw on themes from international digital-governance benchmarks such as the OECD Digital Government Index and the World Bank GovTech Maturity Framework [4,5,9].

7.3 Institutional and Legal Integration

For F-Procure or similar architectures to achieve formal recognition, alignment with existing procurement legislation and administrative practice is essential. Governments considering pilots could:

- Update procurement manuals and e-tender regulations to acknowledge blockchain-based records and smart-contract-driven workflows, using model clauses that recognise “electronic distributed ledgers” under national evidence law [2,4,11,12,16];
- Engage with national or regional standards bodies (for example, BSI, ISO/TC 307, ITU-T DLT groups) to

ensure that technical choices are consistent with emerging best practice [12,13];

- Define roles and responsibilities for public auditors, inspectorates and anti-corruption agencies through memoranda of understanding, (MoUs), particularly in relation to data access, ethical oversight and dispute resolution [4,16].

These steps would not be implemented unilaterally by the technology itself; rather, they require coordinated policy and legal processes in which F-Procure functions as an enabling infrastructure. The MVP presented in this paper should therefore be understood as an input into such processes rather than as a ready-made solution.

7.4 Risk Management and Mitigation

Adopting decentralised procurement infrastructure introduces new categories of risk that must be explicitly managed:

- **Legal risk:**
Use legislative or regulatory sandboxes to permit controlled smart-contract trials under supervisory oversight, clarifying liability, evidential status and recourse mechanisms in the event of disputes or system failures.
- **Technical risk:**
Implement redundancy through multi-region node deployment, robust key-management practices and, where feasible, formal or semi-formal verification of critical smart-contract logic. Consideration should be given to MEV-related risks and to the resilience of the platform’s consensus and oracle mechanisms.
- **Data-privacy risk:**
Ensure that personal and commercially sensitive data remain encrypted or pseudonymised; conduct data-protection impact assessments and align with GDPR, the UK Data Protection Act 2018 and relevant local legislation [10–12]. Explore privacy-enhancing technologies where appropriate.
- **Governance risk:**
Maintain public oversight through open-data portals (for non-confidential information), independent audits and clear channels for suppliers and citizens to raise concerns or contest outcomes. Governance arrangements should address conflict-of-interest management, change-control processes and incident response.
- **Change-management risk:**
Introduce structured training, phased roll-out and iterative policy updates to prevent institutional disruption and to build confidence among procurement officers, vendors and oversight bodies. Co-design processes involving users and stakeholders can help align technical features with administrative realities.

7.5 Long-Term Policy Outlook

In the longer term, systems such as F-Procure could serve as strategic policy tools for economic inclusion and sustainable development. By embedding measurable indicators for fairness, diversity and supplier participation, blockchain-enabled procurement platforms may support national strategies aligned with the United Nations Sustainable Development Goals – notably Goal 16 (Peace, Justice and Strong Institutions) and Goal 9 (Industry, Innovation and Infrastructure) [9,16].

Realising this potential will require more than technical refinement. It will depend on sustained collaboration between technologists, policymakers, legal scholars, civil-society organisations and the private sector. Issues such as long-term cryptographic resilience, MEV-resilient auction design and cross-border legal interoperability will need continued research and experimentation [2,4,12,13]. The MVP described in this paper should therefore be viewed as a starting point for structured experimentation, rather than as a finished solution, with the policy and adoption roadmap offering a basis for dialogue with governments and international organisations considering pilot deployments.

8. Conclusion and Future Research Directions

This study has explored the design and MVP implementation of F-Procure, a decentralised, modular procurement framework built around five interrelated layers: blockchain execution, smart contracts, ERP interoperability, AI analytics and compliance-aware governance. Using small-scale, manually executed synthetic procurement scenarios on Flare's Coston2 and Songbird test networks, the research shows that core procurement workflows—tender creation, commit–reveal bidding, award and escrow—can be modelled and executed in a way that produces transparent, verifiable audit trails under controlled conditions.

The findings provide indicative evidence that a modular, smart-contract-based architecture can reduce manual reconciliation effort in bid evaluation and audit preparation in the specific scenarios examined. The experiments also suggest that selected procedural requirements, such as deadline enforcement and basic documentation checks, can be expressed as programmable conditions and enforced consistently. At the same time, the work does not claim statistically generalisable efficiency gains, comprehensive “compliance-by-code” for any particular legal framework or scalability to high-volume public procurement. Those remain open questions for future empirical research and piloting.

The study underscores the distinction between technical feasibility and institutional readiness. The F-Procure architecture indicates that integration with ERP systems, AI-assisted oversight and governance mechanisms is technically plausible, but these components are only partially realised in the MVP and have not yet been tested with real tenders or stakeholders. Successful deployment will depend on regulatory recognition of distributed

ledgers, appropriate evidence rules, staff training and the development of governance structures capable of overseeing algorithmic tools, managing change and handling incidents.

8.1 Limitations

This work has several limitations which should be acknowledged explicitly. First, the empirical evaluation is restricted to small numbers of manually executed synthetic procurement scenarios on test networks, and therefore cannot support statistically robust claims about scalability, throughput, transaction costs or long-run organisational impact. Secondly, only parts of the proposed five-layer architecture are fully implemented in the current MVP: the blockchain and smart-contract layers are exercised end-to-end, whereas the ERP interoperability, AI analytics and compliance layers are realised in simplified or conceptual form and have not yet been integrated with live systems or real tenders. Thirdly, the implementation and experiments are conducted on a single technical stack, using Flare as a testbed; although the architectural ideas are intended to be platform-agnostic, no comparative analysis with alternative DLT or oracle frameworks is undertaken here. Finally, while the paper draws on procurement and data-protection frameworks for conceptual alignment, it does not constitute a formal legal validation of any jurisdiction's procurement regime, nor does it address all emerging issues in depth, including MEV-resilient auction design, advanced privacy techniques and post-quantum security. These limitations delineate the scope of the contribution and motivate the future research directions outlined below.

8.2 Future Research Directions

From a research perspective, several directions merit further exploration:

- **Larger-scale empirical evaluation:** Automated simulations and, ultimately, pilots with real procurement datasets are needed to assess scalability, throughput, transaction costs, error rates and energy characteristics under realistic load. This includes more rigorous comparisons with existing e-procurement platforms and with alternative blockchain or DLT infrastructures.
- **Richer technical artefacts and reproducibility:** Future work should provide more detailed technical artefacts—such as public repositories for non-sensitive smart-contract code, contract interfaces, example transaction hashes and reference implementations of the ERP connector—to strengthen reproducibility and independent scrutiny.
- **Comparative and platform-agnostic analysis:** While Flare serves as a useful testbed in this MVP, further research could compare F-Procure's architectural principles when implemented on different platforms and oracle frameworks. Such comparisons would help clarify which aspects of the

framework are platform-specific and which are broadly applicable across the DLT ecosystem.

- **Advanced privacy, integrity and security mechanisms:**

Building on the basic commit–reveal structure, future work may examine MEV-resilient auction designs, privacy-enhancing technologies (such as zero-knowledge proofs) and post-quantum security considerations, in collaboration with cryptographers, economists and legal experts.

- **AI governance and ethics in procurement:**

The minimal anomaly-detection logic used in this MVP points towards a broader research agenda on AI-assisted procurement oversight. Future studies should investigate explainability, fairness and accountability in algorithmic assessments and explore governance frameworks for the responsible use of AI in public contracting.

The next stage of the F-Procure roadmap envisages collaboration with public-sector bodies and regional development agencies to design carefully scoped pilots. These pilots would test the framework under real procurement conditions, focusing on institutional readiness, legal interoperability and socio-economic outcomes such as SME participation and supplier diversity. They would also provide an opportunity to refine governance structures, compliance templates and KPI frameworks in partnership with oversight institutions and civil-society stakeholders. As illustrated in Figure 10, the intended trajectory runs from research and MVP experimentation through pilot deployment and evaluation, with governance shaping the eventual emergence of regulation and institutional adoption.

Ultimately, F-Procure is not presented as a finished solution, but as a research-driven contribution to the broader discourse on digital governance and public procurement reform. Its modular architecture and MVP implementation are intended to serve as a reference point for further experimentation, peer review and adaptation across jurisdictions. As blockchain and related technologies mature, frameworks of this kind may help shape a class of digital public infrastructure in which trust is supported by verifiable processes and transparent governance, rather than by retrospective audits alone.

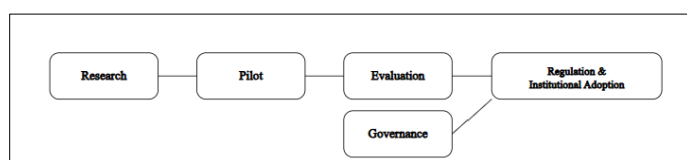


Figure 10. Research–policy integration pathway (conceptual). Conceptual illustration of how design-science research (such as the F-Procure MVP) can inform pilot projects, which in turn feed into policy development, standard-setting and longer-term institutional adoption. The pathway represents a proposed trajectory for future work rather than steps already undertaken in this study.

Competing Interests:

The author declares that they have no competing financial interests or personal relationships that might be perceived as affecting the impartiality of this work.

Ethical approval:

Not applicable. This study involves system design and technical evaluation only and does not involve human or animal subjects.

Author's contribution:

Conceptualisation, system design, software development, writing and editing were carried out by “the author”, who was solely responsible for all stages of research execution and manuscript preparation.

Funding:

No external funding was received for this study. Research and prototype development were conducted under an independent innovation programme of the development team.

Acknowledgements:

Prototype development and applied research were undertaken by the development team. The author gratefully acknowledges the Flare Network community for openly accessible technical documentation and support resources, which informed aspects of the system design.

References:

- [1] Deloitte; European Commission, *Study on Up-Take of Emerging Technologies in Public Procurement – D01.06 Final Report: Emerging Technologies in Public Procurement around the World*. Luxembourg: Publications Office of the European Union, 2020.
- [2] D. Allesie, M. Sobolewski, and L. Vaccari, *Blockchain for Digital Government: An Assessment of Pioneering Implementations in Public Services*. Luxembourg: Publications Office of the European Union, 2019.
- [3] World Economic Forum, *Exploring Blockchain Technology for Government Transparency: Blockchain-based Public Procurement to Reduce Corruption*. Geneva: World Economic Forum, 2020. Supplementary research report. Available: https://www3.weforum.org/docs/WEF_Blockchain_Government_Transparency_Report_Supplementary%20Research.pdf.
- [4] Organisation for Economic Co-operation and Development (OECD), *Public Procurement and Infrastructure Governance: OECD Recommendations*. Paris: OECD Publishing, 2020.
- [5] World Bank, *GovTech Maturity Index: The State of Public Sector Digital Transformation*. Washington (DC): World Bank, 2021.
- [6] C. Ferreira, J. Almeida, and R. Pereira, “Smart contracts in public procurement: Opportunities and challenges,” *Gov. Inf. Q.*, vol. 38, no. 4, pp. 101612, 2021.
- [7] Flare Networks, Flare documentation [Internet]. 2025 [cited 2025 Nov 3]. Available: <https://docs.flare.network/>.
- [8] Chainlink Labs, *Chainlink 2.0: Next Steps in the Evolution of Decentralized Oracle Networks*. Chainlink Research

Whitepaper [Internet]. 2021 [cited 2025 Nov 3]. Available: <https://research.chain.link/whitepaper-v2.pdf>.

- [9] Organisation for Economic Co-operation and Development (OECD), *Convention on Combating Bribery of Foreign Public Officials in International Business Transactions*. Paris: OECD Publishing, 2011.
- [10] United Nations Department of Economic and Social Affairs, *Sustainable Development Goals: SDG 9 and SDG 16* [Internet]. New York: United Nations, 2023 [cited 2025 Nov 3]. Available: <https://sdgs.un.org/goals>.
- [11] European Union, “Directive 2014/24/EU of the European Parliament and of the Council of 26 February 2014 on public procurement,” *Official Journal of the European Union*, vol. L94, pp. 65–242, 2014.
- [12] UK Parliament, *Data Protection Act 2018*. c.12. London: The Stationery Office, 2018.
- [13] ITU-T Focus Group on Application of Distributed Ledger Technology, *Distributed Ledger Technology Standardization Landscape*. Geneva: International Telecommunication Union, 2019. (FG DLT D1.3).
- [14] A. R. Hevner, S. T. March, J. Park, and S. Ram, “Design science in information systems research,” *MIS Q.*, vol. 28, no. 1, pp. 75–105, 2004.
- [15] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A design science research methodology for information systems research,” *J. Manag. Inf. Syst.*, vol. 24, no. 3, pp. 45–77, 2007.
- [16] British Standards Institution, *Distributed Ledger Technologies – Governance and Interoperability*. PAS 3333:2023. London: BSI, 2023.